

Can a Neighborhood Become Louder on Paper?

A sensitivity sketch for Greenpoint's 2015 311 noise archive

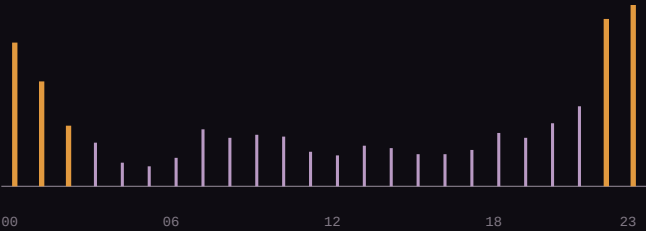
Observed map - 1,031 coordinate pairs - filled points sized by request count



Lavender 2,393 rows inside - 1.62 requests per plotted location

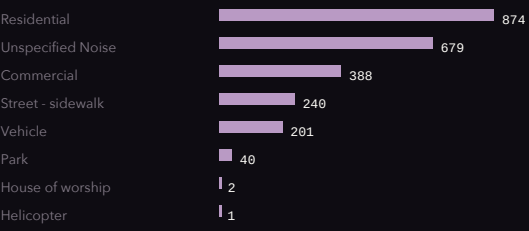
Orange rings 6 edge locations - within 1.2 m of proxy boundary
Record counts 24 + 4 + 1 + 1 + 1 + 1 + 1 = 32 rows

Observed - records by created hour 23:00 = 299



22:00-02:59 = 1,085 records - 44.7% of the file

Observed - all 8 complaint types



01 What I am attempting

Can Greenpoint become louder in the municipal record without becoming louder in reality? I hold the acoustic environment conceptually constant and change only reporting probability. This matters because complaint volume can be mistaken for acoustic conditions.

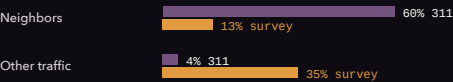
02 The dataset

NYC311 recorded these requests and routed them to responding agencies. The archive documents non-emergency service requests directed to specific agencies. Filter: 2015; incident ZIP 11222; complaint type begins "Noise"; valid latitude/longitude. Result: 2,425 geocoded rows, 0 duplicate keys and 8 complaint types.

03 Research

DOHMH shows that complaint sources and reported sleep disturbance are not equivalent. Kontokosta and Hong document socio-spatial disparities in 311 reporting in Kansas City. Gong et al. use Poisson models for NYC noise-complaint counts.

Published comparison - NYC-wide



DOHMH Brief 105 - NYC study dates, 2016-17

04 My record

A source row is one 2015 service request, not one sound. I aggregate the file into 192 type-hour baseline cells (8 types x 24 hours); source rows remain a location library. Fields: ID, date-time, categorical type and descriptor, and latitude/longitude in decimal degrees.

Actual row - not synthetic

29607573 - 2015-01-01 00:28
Residential - loud music/party

File audit

1,031 unique coordinate pairs
Jan 1-Dec 31 - API total rechecked: 2,425

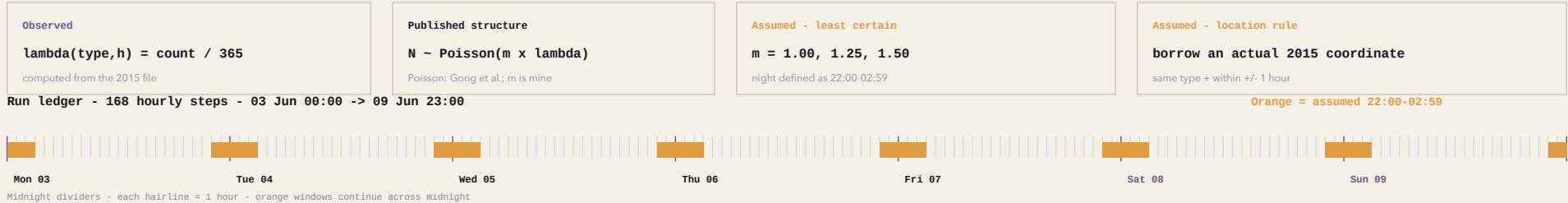
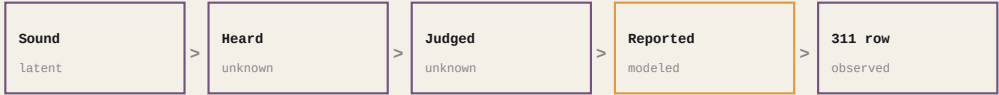
The Model Does Not Forecast Sound.

It tests how a change in reporting could change the city's record.

05 My rule

At each simulated hour, every type-hour cell supplies $\lambda = \text{count} / 365$. A Poisson draw creates zero or more request rows; each row borrows a real coordinate from the same type within ± 1 hour. Only the reporting multiplier changes.

Where a sound becomes data



06 My run

From 2030-06-03 00:00 to 2030-06-09 23:00: 168 hourly steps, repeated 100 times to show simulation spread. The future dates, repeat count and higher multipliers are design choices; Poisson counts and borrowed locations are random.

07 The data it would generate

Each simulated request would add a time, complaint type, borrowed source key and coordinate, scenario and multiplier. Outputs: hourly count bands and two point maps. No future result is claimed on this page.

08 What it cannot see

Decibels, duration, unreported noise, repeated callers, access to 311, caller identity, new venues, population change or why a sound becomes unacceptable. More rows do not necessarily mean more sound.

Output schema - three worked examples

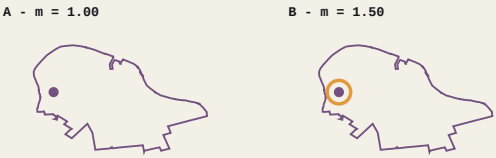
future time	type from source	source key	source lat/lon	scenario	m
2030-06-03 22:00	Commercial	29631631	40.729797, -73.958385	A	1.00
2030-06-04 23:00	Residential	29626818	40.719768, -73.948692	B1	1.25
2030-06-05 00:00	Residential	29687573	40.721473, -73.946979	B2	1.50

Type, source key and coordinate are real 2015 rows; future time, scenario and m are assigned. Not forecast results.

Sources - exact scope

NYC Open Data, 311 Service Requests 2010-2019, 76ig-c548; filter stated on p.1. MODZCTA boundary: pri4-ifjk.
Teoh & Olson, NYC DOHMH Epi Data Brief 105 (2018). Kontokosta & Hong, SCS 64 (2021), 102503.
Gong et al., Urban Informatics 4 (2025), 24, doi:10.1007/s44212-025-00093-9. API total rechecked 29 Sep 2026.

Map output - one symbol example



Both maps use real source row 29631631.
Orange ring = a possible additional B record.
Symbol example only; no future count was computed.